Algorithms and Programming

NAVIGAZIONE QUIZ

1 2 3 4 5 6

Successivo

Tempo rimasto **1:49:37**

Avvia una nuova anteprima

NAVIGAZIONE

Dashboard
- Home del sito
- Pagine del sito
- Corso in uso
- I miei corsi

AMMINISTRAZIONE
- Gestione Quiz
- Amministrazione del corso
- Cambia ruolo in ...

Puoi visualizzare l'anteprima del quiz, tuttavia se questo fosse un tentativo reale, non ti sarà possibile in quanto:

Spiacente, il quiz non è disponibile

Domanda 1
Risposta non ancora data
Punteggio max.: 1,00
Contrassegna domanda
Modifica domanda

The following function myf is called with the string in storing **"This is a very loooong string"**, and with the string out and the value n undefined.

```
void myf (char *in, char *out, int *n) {
  char *tmp1, *tmp2;
  int l;
  out[0] = '\0';
  tmp1 = in;
  while (*tmp1!='\0') {
    while (*tmp1==' ') {
      tmp1++;
    }
    tmp2 = tmp1;
    while (*tmp2!=' ' && *tmp2!='\0') {
      tmp2++;
    }
    l = tmp2 - tmp1;
    if (l > strlen(out)) {
      *n=l;
      strncpy (out, tmp1, l);
      out[l] = '\0';
    }
    tmp1=tmp2;
  }
  return;
}
```

Specify the value of the string **out** and the integer **n** returned by the function.

Please, report out and n on the same line separated by one single space. No other symbols must be included in the response. This is an example of the response: abcd 12

Risposta:

Domanda 2
Risposta non ancora data
Punteggio max.: 1,00
Contrassegna domanda
Modifica domanda

The following recursive function is called with the string s storing **"This is my string !"**

```
int rec (char *s) {
  if (*s=='\0')
    return 0;
  else
    return (((*s==' ') ? 1 : 0) + rec(s+1));
}
```

Indicate the exact value returned by the function.

Please, report a single integer value. No other symbols must be included in the response. This is an example of the response: 0

Risposta:

Domanda 3
Risposta non ancora data
Punteggio max.: 1,00
Contrassegna domanda
Modifica domanda

The following recursive function is called with n=4.

```
int rec (int n) {
  int i, res;
  if (n==0)
    return (1);
  res = 0;
  for (i=0; i<n; i++)
    res += rec(i) * rec(n - 1 - i);
  return res;
}
```

Following the recursion tree, report the exact value returned by the function to the initial caller.

Please, report the exact output displayed by the function. No other symbols must be included in the response. This is an example of the response: 0

Risposta:

Domanda 4
Risposta non ancora data
Punteggio max.: 1,00
Contrassegna domanda
Modifica domanda

A file includes an undefined number of rows. Each row stores a string and a sequence of integer values. The first integer for each row indicates the number of values following it on the same row. An example of such a file is reported in the following:

```
stringA 5 2 3 4 5 6
stringB 2 7 8
stringC 4 9 8 7 6
```

The function **myf** stores the file into a list of lists. The main list stores the strings. The secondary lists store the integer values.

The C data structures required to define the list of lists and function **myf** are the following:

```
typedef struct e1_s e1_t;
typedef struct e2_s e2_t;
struct e1_s {
  char *str;
  e2_t *right;
  e1_t *next;
};
struct e2_s {
  int val;
  e2_t *next;
};
```

```
void myf (char *name) {
  FILE *fp;
  char str[100];
  int i, n;
  e1_t *p1, *head = NULL;
  e2_t *p2;
  fp = fopen (name, "r");
  while (fscanf (fp, "%s", str) != EOF) {
    p1 = malloc (1 * sizeof (e1_t));
    p1->str = strdup (str);
    p1->right = NULL;
    p1->next = head;
    head = p1;
    fscanf (fp, "%d", &n);
    for (i=0; i<n; i++) {
      p2 = malloc (1 * sizeof (e2_t));
      fscanf (fp, "%d", &(p2->val));
      // Instruction
    }
  }
  fclose (fp);
  return;
}
```

Select the right set of instructions which must substitute the the line

// Instruction

such that the list of lists is correctly created.

- (a) p2->next = p1->right; p1->next = p2;
- (b) p1->next = p2; p1->right = p2->next;
- (c) p1->next = p2->next; p1->right = p2;
- (d) p2->next = p1->right; p1->right = p2->next;
- (e) p2->next = p1->right; p1->right = p2;

Domanda 5
Risposta non ancora data
Punteggio max.: 4,00
Contrassegna domanda
Modifica domanda

A file specifies an electric board of rectangular size. The board can be represented by a rectangle matrix in which each element corresponds to a C pointer to a string storing the name of the electric element occupying that area. The file specifies the number of row R and column C on the first row (integer values). On all following rows, it specifies a row index, a column index, and a string, i.e., the name of the component in that specific position. Strings have undefined length but have at most 50 characters. The following is a correct example of such a file.

```
5 10
0 2 resistor
0 3 diode
3 4 transistor
2 1 inductor
etc.
```

Write the function **board_read** which receives the file name as a parameter, it opens the file, allocates all dynamic structures required to store the file with the specified format, and it returns the pointer to such a data structure. When an element of the matrix does not have a corresponding component it has to be equal to the NULL pointer.

The prototype of the function is the following:

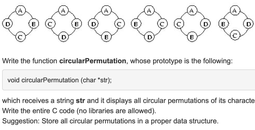
```
char ***board_read (char *name);
```

Please, report the entire C code.

Domanda 6
Risposta non ancora data
Punteggio max.: 6,00
Contrassegna domanda
Modifica domanda

The number of ways to arrange n distinct objects along a straight line is given by the **permutation**. The number of permutation of n objects is *n!*.
The number of ways to arrange n distinct objects along a circular line is given by the **circular permutation**. The number of circular permutations is *(n-1)!*. The number is *(n-1)!* since all cyclic permutations of objects are equivalent because the circle can be rotated.

For example, of the 3!=6 permutations of three objects {A,B,C}, the (3-1)!=2 distinct circular permutations are {A,B,C} and {A,C,B} as represented in the top section of the following figure. The other permutations, i.e., {B,C,A}, {C,A,B}, {C,B,A}, and {B,A,C}, are equivalent because they cannot be distinguished by the previous two when placed around a circle.
Similarly, of the 4!=24 permutations of four objects, only (4-1)!=6 are distinct circular permutations as represented by the bottom part of the figure.



Write the function **circularPermutation**, whose prototype is the following:

```
void circularPermutation (char *str);
```

which receives a string **str** and it displays all *circular* permutations of its characters.
Write the entire C code (no libraries are allowed).
Suggestion: Store all circular permutations in a proper data structure.

Successivo